Definition 4.2 (Odd and Even functions)

A real valued function f is an **even function** if for all x in the domain of f, $-x$ is also in the domain of f and $f(-x) = f(x)$.

A real valued function f is an **odd function** if for all x in the domain of f, $-x$ is also in the domain of f and $f(-x) = -f(x)$.

For instance, x^3, $\sin x$, $\operatorname{cosec} x$, $\tan x$ and $\cot x$ are all **odd functions**, whereas x^2, $\cos x$ and $\sec x$ are **even functions**.

Remark

The period of $f = g \pm h$ is $\operatorname{lcm}\{\text{period of } g, \text{period of } h\}$, whenever they exist.

For instance, the period of $y = \cos 6x + \sin 4x$ is π and that of $y = \cos x - \sin x$ is 2π.

4.2.1 Domain and Range of trigonometric functions

The **domain** and **range** of trigonometric functions are given in the following table.

Trigonometric function	$\sin x$	$\cos x$	$\tan x$	$\operatorname{cosec} x$	$\sec x$	$\cot x$
Domain	$\mathbb{R}$	$\mathbb{R}$	$\mathbb{R} \setminus \left\{(2n+1)\frac{\pi}{2}, n \in \mathbb{Z}\right\}$	$\mathbb{R} \setminus \{n\pi,\ n \in \mathbb{Z}\}$	$\mathbb{R} \setminus \left\{(2n+1)\frac{\pi}{2}, n \in \mathbb{Z}\right\}$	$\mathbb{R} \setminus \{n\pi,\ n \in \mathbb{Z}\}$
Range	$[-1, 1]$	$[-1, 1]$	$\mathbb{R}$	$\mathbb{R} \setminus (-1, 1)$	$\mathbb{R} \setminus (-1, 1)$	$\mathbb{R}$

4.2.2 Graphs of functions

Let $f : \mathbb{R} \to \mathbb{R}$ be a real valued function and $f(x)$ be the value of the function f at a point x in the domain. Then, the set of all points $(x, f(x)), x \in \mathbb{R}$ determines the graph of the function f. In general, a graph in xy-plane need not represent a function. However, if the graph passes the vertical line test (any vertical line intersects the graph, if it does, atmost at one point), then the graph represents a function. A best way to study a function is to draw its graph and analyse its properties through the graph.

Every day, we come across many phenomena like tides, day or night cycle, which involve periodicity over time. Since trigonometric functions are periodic, such phenomena can be studied through trigonometric functions. Making a visual representation of a trigonometric function, in the form of a graph, can help us to analyse the properties of phenomena involving periodicities.

To graph the trigonometric functions in the xy-plane, we use the symbol x for the independent variable representing an angle measure in radians, and y for the dependent variable. We write $y = \sin x$ to represent the sine function, and in a similar way for other trigonometric functions. In the following sections, we discuss how to draw the graphs of trigonometric functions and inverse trigonometric functions and study their properties.

4.2.3 Amplitude and Period of a graph

The amplitude is the maximum distance of the graph from the x-axis. Thus, the amplitude of a function is the height from the x-axis to its maximum or minimum. The period is the distance required for the function to complete one full cycle.

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